UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Classes 003–004 · Total derivative, chain rule and implicit functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: "Total derivative, chain rule and implicit functions"
T003b 00:01:10	Transcript 003 part II: "Problems on total derivatives, chain rule"
T004a 00:13:09	Transcript 004 part I: "Implicit functions and problems"
T004b 00:02:31	Transcript 004 part II: "Problems on composite and implicit functions"
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question

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1 Composite functions and the one-variable chain rule

Definitions

Composite function. A function of a function: $f(g(h(x)))$, or a function of several variables whose variables are themselves functions of other variables. S003·4 · T003a 00:00:59

Key points

1. For one variable you differentiate a composite function from the outside in: f first, then g, then h. ESA ★ T003a 00:01:19

2. For several variables the first question is always “differentiate with respect to what?”, because there is more than one variable. ESA ★ T003a 00:01:45

2 The total derivative rule

Definitions

Total derivative. If $z = f(x, y)$ where x and y are functions of one independent variable t, the ordinary derivative dz/dt is the total derivative of z with respect to t. S003·5 · T003b 00:00:52

Tree diagram. A diagram with z at the root, the intermediate variables in the middle and the final variable(s) at the leaves. Each branch is labelled with a derivative; multiply along each path and add over the paths. S003·5 · S003·14 · T003a 00:19:22 · T003a 00:25:02

Statements in the class material

> *"If $z = f(x, y)$ where x and y are functions of one independent variable 't' then we can find the derivative of z with respect t as* $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$*."* S003·5

Key points

1. You can always substitute $x(t)$ and $y(t)$ into z and differentiate directly, but this is often messy. The total derivative rule avoids substituting first. ESA ★★ T003a 00:02:39 · T003a 00:03:13
2. Both methods must give the same answer; substituting first is a good check on a short example. ESA ★★ (added for clarity) T003a 00:04:35
3. After using the rule, write the answer in terms of t alone (substitute $x(t)$, $y(t)$), or evaluate at the given t. ESA ★★ S003·19 · T003b 00:02:03
4. At a given t, a term vanishes if its dx/dt or dy/dt is 0 there (e.g. $dy/dt = \cos t = 0$ at $t = \pi/2$). ESA ★★ T003b 00:02:34

Why?

Why is it dz/dt (ordinary) and not $\partial z/\partial t$?
After substituting $x(t)$ and $y(t)$, z depends on the single variable t, so its derivative with respect to t is an ordinary derivative. The partial signs belong only to z with respect to x and y, where the other variable is held fixed.
S003·5 · T003a 00:02:23

Exam traps

Exam trap. Slide 003·6 has wrong numbers. For $z = x^2y^3$, $x = t^2$, $y = 2t^3$: $z_x = 2xy^3 = 16t^{11}$ (not $2t^{11}$), $z_y = 3x^2y^2 = 12t^{10}$ (not $9t^{10}$), and $dz/dt = 16t^{11}(2t) + 12t^{10}(6t^2) = 104t^{12}$ (not $60t^{12}$). Check: $z = 8t^{13}$. (added for clarity) S003·6

3 The total differential: approximation and sensitivity

Definitions

Total differential. For $z = f(x, y)$: $dz = f_x\,dx + f_y\,dy$. For $z = f(x_1, \ldots, x_n)$: $dz = f_{x_1}dx_1 + \cdots + f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

Statements in the class material

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

Key points

1. The total differential estimates how much f changes from a known point to a nearby one. ESA ★★ S003·8
2. Sensitivity: in $dV = 40\pi\,dr + 4\pi\,dh$ the coefficient of dr is ten times that of dh, so the tank's volume is more sensitive to the radius than to the height. ESA ★★ S003·11 · S003·12 · T003a 00:18:30
3. Sometimes you need not finish the problem: the coefficients of the differential already tell you which variable contributes most. ESA ★ T003a 00:19:05
4. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ (transcript only) T004a 00:07:57 · T004a 00:08:29
5. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ (transcript only) T004a 00:08:47

Why?

Why does the tank's volume depend more on the radius?
$V = \pi r^2 h$ has $V_r = 2\pi r h = 40\pi$ and $V_h = \pi r^2 = 4\pi$ at $r = 2$, $h = 10$. So $dV = 4\pi(10\,dr + dh)$: a change in r is multiplied by ten times as much as the same change in h.
S003·11 · T003a 00:18:45

Exam traps

Exam trap. Use signed changes: from -3 to -3.03, $dy = -0.03$ (slide 003·8 writes 0.03 but correctly uses -0.03). S003·8

4 Rates of change problems

Definitions

Rate of change. The derivative of a quantity with respect to time t; positive means increasing, negative means decreasing. T003a 00:12:06

Key points

1. If $dz/dt > 0$ the quantity is increasing at that instant; if $dz/dt < 0$ it is decreasing. ESA ★★ T003a 00:12:06
2. When an unknown rate such as dy/dt is needed, find it from the constraint by implicit differentiation, e.g. $x^2 + y^2 = 5 \Rightarrow dy/dt = -\frac{x}{y}\,dx/dt$. ESA ★★ S003·7 · T003a 00:09:29
3. A decreasing side has a negative rate: "decreasing at 2 cm/s" means $dy/dt = -2$. ESA ★★ S003·10

Why?

Why does the angle have to decrease in the triangle problem?
$A = \frac{1}{2}xy\sin\theta$. One side grows (+3) and the other shrinks (−2), but at the given instant their combined effect $\frac{1}{2}\sin\theta(3y - 2x) = \frac{1}{2}\cdot\frac{1}{2}(90 - 40) > 0$ would increase the area. To keep A constant, θ must decrease.

ADDED FOR CLARITY S003·9 · S003·10

Exam traps

Exam trap. Slide 003·10 prints $d\theta/dt = 0.04831$. The value is $-25/(300\sqrt{3}) = -\sqrt{3}/36 \approx -0.0481$ rad/s: negative, and 481, not 4831. ADDED FOR CLARITY S003·10 · T003a 00:16:55

Exam trap. $x^2/16 - y^2/9 = z$ is a hyperbolic paraboloid (a saddle), not a cup-shaped paraboloid. The calculation does not change. ADDED FOR CLARITY S003·7 · T003a 00:06:27

5 The chain rule for composite functions

Definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

Statements in the class material

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Key points

1. When x and y depend on two variables u, v, z cannot be differentiated "totally": only z_u and z_v exist. ESA ★★ T003a 00:20:40
2. Write the tree diagram every time a function has more than one independent variable, so that no partial derivative is missed. ESA ★★★ T003a 00:25:02
3. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
4. If f is not given ($u = f(x, y)$), keep f_x, f_y (or f_r, f_θ) as symbols in the answer. ESA ★★★ S003·17 · S003·25 · T003a 00:26:28
5. For $f(2x - 3y,\ 3y - 4z,\ 4z - 2x)$, name the arguments r, s, t and compute $r_x, s_x, t_x, \ldots$; most are constants or 0. ESA ★★★ S003·15 · T003a 00:22:33
6. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28
7. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ (added for clarity) S004·10 · S004·13 · S004·14
8. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ (added for clarity) S004·15
9. The chain rule extends to any number of variables: z of x, y, each of x, y of three variables, and so on. ESA ★ T003a 00:21:40

Why?

Why does $u = f(x - y,\ y - z,\ z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.

(added for clarity) ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2+t^2}$, ...) can.

S003·17 · T003a 00:26:28

Exam traps

Exam trap. Slide 003·16 shows $\dfrac{\partial v}{\partial y} = -3\dfrac{\partial v}{\partial r} + \dfrac{\partial v}{\partial s}$; it should be $-3\dfrac{\partial v}{\partial r} + 3\dfrac{\partial v}{\partial s}$ because $s = 3y - 4z$ gives $s_y = 3$. With the 3 the proof works. ADDED FOR CLARITY S003·16

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

Exam trap. Write $\dfrac{\partial z}{\partial u}$, not $\dfrac{dz}{du}$: z depends on both u and v. ADDED FOR CLARITY S003·13

6 Implicit functions of two variables

Definitions

Implicit function. A function given by a relation $f(x,y)=c$ (for example $x^3y+3xy^2=3$) rather than by a formula $y=g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z=f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2+y^2=c$ and the spheres $x^2+y^2+z^2=c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

Statements in the class material

> *"If z represents a level surface, i.e. if $z=f(x,y)=c$, then $dz=0$. The total derivative is also called the differential."* S004·4

> *"Setting $\dfrac{dz}{dx}=0$ leads to a simple computation formula for implicit differentiation: $f_x\cdot 1+f_y\dfrac{dy}{dx}=0$."* S004·5

Key points

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx=-f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. Restriction to remember: the formula needs $z=f(x,y)$ to be a constant (a level curve). Move everything to one side first: $f(x,y)=c$ or $f(x,y)=0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
5. The slope dy/dx at a point gives the tangent line $y-y_0=m(x-x_0)$. ESA ★★ S004·7 · T004a 00:12:01
6. For $(\cos x)^y=(\sin y)^x$ the result needs $\cos x>0$ and $\sin y>0$, because of the logarithms. ESA ★ T004a 00:18:12
7. Implicit and composite together: if $u=\tan(x^2+y^2)$ and $x^2-y^2=2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why?

Why is $dy/dx=-f_x/f_y$?
Along the curve $f(x,y)=c$ the value of f does not change, so $df=f_x\,dx+f_y\,dy=0$. Divide by dx: $f_x+f_y\dfrac{dy}{dx}=0$. The branch diagram (Figure G6.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2, 1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.

ADDED FOR CLARITY S004·4 · S004·5

Exam traps

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x, y)/f_x(x, y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln \cos x$). S004·8 · T004a 00:16:02

7 Implicit functions of three variables

Definitions

Implicit function of three variables. $f(x,y,z)=0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

Statements in the class material

> *"Let $f(x,y,z)=0$ be the equation of an implicit function of three variables x,y,z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Key points

1. For $f(x,y,z)=0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2+3xyz+\sin(e^z)=c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f=0$ before substituting. ESA ★★ (added for clarity) S004·9 · T004a 00:18:39

Exam traps

Exam trap. The lecture works problem 3 at $(1,0,2)$ and gets 0 for both derivatives, but $(1,0,2)$ is not on $z^3+xy-y^2z=6$ (it gives 8). Use the slide's point $(0,1,2)$: $\partial y/\partial x=1/4$, $\partial z/\partial x=-1/11$. (added for clarity) S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W7.3). (added for clarity) ESA Jan-24 s3 1d

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In-scope Q1 parts: five chain-rule questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks each; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks each) and one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks). The total derivative and rates of change were not asked in these papers; they are computed in class-work problems 9–13. No paper repeats another. Questions on Euler's theorem, Taylor series, maxima and Lagrange belong to later sessions and are left out.

5 · The chain rule for composite functions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W5.11

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q1.d · solved in W5.12

If $u = f(x - y,\, y - z,\, z - x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.b · solved in W5.13

If $z = f(u, v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2 + v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.b · solved in W5.14

If $z = f(x, y)$ and $x = e^u \sin v$, $y = e^u \cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 · 6 marks · UE23MA141A Q1.a · solved in W5.15

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.

7 · Implicit functions of three variables

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.d · solved in W7.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x \log(ex)]^{-1}$.